

DAEN 427 Homework 2

Due date: Aug 26, 2025

1. From the following expressions, which one corresponds to the sentence: *The probability of being sunny given that it is June 14th of 1976?*

- (a) $p(\text{sunny})$
- (b) $p(\text{sunny} \mid \text{June})$
- (c) $p(\text{sunny} \mid \text{June 14th of 1976})$
- (d) $p(\text{June 14th of 1976} \mid \text{sunny})$
- (e) $p(\text{sunny, June 14th of 1976})/p(\text{June 14th of 1976})$

Answer: The correct answer are **c** and **e**. (the last one follows from the definition of conditional probability)

2. Show that the probability of choosing a human at random and picking the Pope is not the same as the probability of the Pope being human. In the animated series Futurama, the (Space) Pope is a reptile. How does this change your previous calculations?

Answer: Let's assume there are $7.8 \cdot 10^8$ humans and there is only 1 Pope (Pope Leo XIV at the time of this writing). If a human is picked at random from the entire human population, the chances of that human being the pope are 1 in $7.8 \cdot 10^8$.

$$p(\text{Pope} \mid \text{human}) = \frac{1}{7.8 \cdot 10^8}$$

Because Pope are selected among humans. We can assert that given someone is the Pope, there is 100% chance he is a human.

$$p(\text{human} \mid \text{Pope}) = 1$$

Regarding the animated series Futurama and the (space) Pope is a reptile. We then have that

$$p(\text{Pope} \mid \text{human}) = 0$$

And

$$p(\text{human} \mid \text{Pope}) = 0$$

3. Sketch what the distribution of possible observed values could be for the following cases:

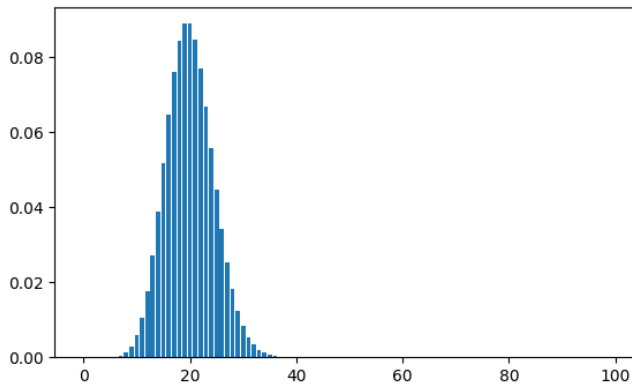
- (a) The number of people visiting your local cafe assuming Poisson distribution.
- (b) The weight of adult dogs in kilograms assuming a Uniform distribution.
- (c) The weight of adult elephants in kilograms assuming Normal distribution.
- (d) The weight of adult humans in pounds assuming skew Normal distribution.

Answer: I'll generate "sketches" using scipy for ease of use presentation in this solution manual. Students make hand sketches please.

- (a) The number of people visiting your local cafe assuming Poisson distribution.

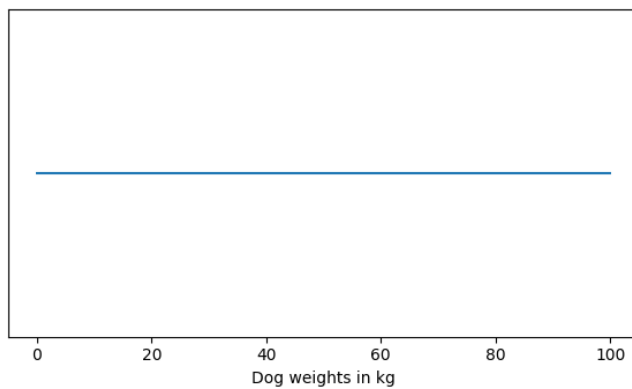
```
# pip install numpy
# pip install scipy
# pip install matplotlib
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

visitors = np.arange(0, 100)
pmf = stats.poisson(20).pmf(visitors)
plt.bar(visitors, pmf)
```



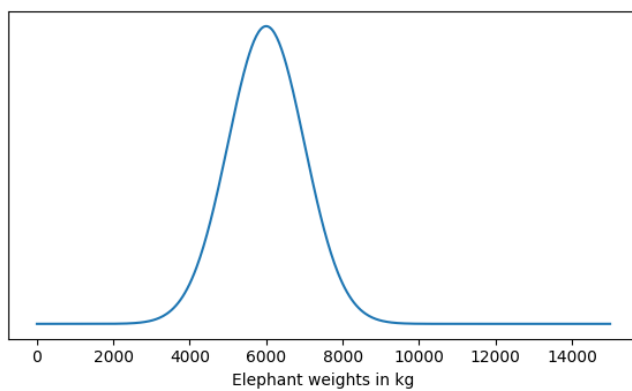
(b) The weight of adult dogs in kilograms assuming a Uniform distribution.

```
dog_weights = np.linspace(0, 100)
# dont need a PDF for this one bc its a uniform distribution
pdf = np.ones(dog_weights.shape[0])
plt.plot(dog_weights, pdf)
plt.yticks([])
plt.xlabel("Dog weights in kg")
```



(c) The weight of adult elephants in kilograms assuming Normal distribution.

```
elephant_weights = np.linspace(0, 15000, 500)
pdf = stats.norm(6000, 1000).pdf(elephant_weights)
plt.clf()
plt.plot(elephant_weights, pdf)
plt.yticks([])
plt.xlabel("Elephant weights in kg")
```



(d) The weight of adult humans in pounds assuming skew Normal distribution.

```
human_weights = np.linspace(0,400, num=1000)
pdf = stats.skewnorm(loc=120, scale=30, a=4).pdf(human_weights)
plt.clf()
plt.plot(human_weights, pdf)
plt.yticks([])
plt.xlabel("Human weights in pounds")
```

